

IoT Oxygen Concentrator Integration

Purity Monitoring, Alert System and Compressor Control

Phase 1 of 4 | System Architecture and Design Reference | Rev 1.0 | June 2026

Phase 1 Architecture	Phase 2 Pictorial	Phase 3 Schematic	Phase 4 PCB
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Current phase highlighted in teal. Document grows with each phase delivered.

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1. Project Overview

This document describes the design and implementation of an IoT integration system for an existing PSA (Pressure Swing Adsorption) oxygen concentrator. The concentrator itself is a commercially acquired, working medical device. No modifications are made to its internal components. Instead, an external integration module is built and housed in a separate enclosure, connected to the concentrator via tubes (for oxygen sampling) and wires (for power and compressor control).

What the integration system does:

Measures oxygen purity: Two OCS-3FL2.0 ultrasonic sensors are placed before and after the filtration stage. This gives a real-time comparison of raw and purified oxygen concentration.

Provides visual feedback: A 0.96-inch OLED display shows live readings from both sensors, system status, GSM signal strength, and system uptime.

Alerts the caregiver: If purity drops below the first threshold (85%), the system sounds a buzzer and sends an SMS to a registered caregiver number. If purity falls further below the second threshold (70%), the buzzer pattern becomes more persistent and the system places a voice call to the caregiver instead.

Controls the compressor: A relay circuit, driven by an optocoupler and transistor, can cut power to the concentrator compressor when oxygen purity is critically low. This protects the patient from receiving dangerously low-purity output.

Operates independently: There is no smartphone app. The system is self-contained. Caregiver contact details are configurable via USB serial (initial setup) or via incoming SMS command (field update).

NOTE: This integration does not modify the internal circuits of the oxygen concentrator. All electrical connections to the concentrator are limited to: (1) the AC mains feed to the compressor relay, and (2) the oxygen outlet tube for sensor sampling.

2. System Architecture

The integration system is built around the ESP32-S3 DevKitC-1 microcontroller, which serves as the central processing unit. All sensors, communication modules, output devices, and control circuitry connect to it. The diagram below describes the high-level data and power flow across the system.

Block-level signal flow:

AC Mains Input (from concentrator internal supply)	12V AC-DC Module Rectified 12V DC output	LM2596 Step-Down Regulated 5V output
--	--	--

From 5V: ESP32-S3, SIM800L (via 1N4001 voltage drop), OLED, Buzzer, Relay driver circuit.

From 12V: Both OCS-3FL2.0 purity sensors (Pin 1 power input).

Data paths:

Signal Path	Interface	Direction	Notes
Sensor 1 (pre-filter) to ESP32-S3	UART1 @ 9600 baud	Sensor to MCU	3.3V TTL, direct connection
Sensor 2 (post-filter) to ESP32-S3	UART2 @ 9600 baud	Sensor to MCU	3.3V TTL, direct connection
SIM800L to ESP32-S3	UART0 @ 9600 baud	Bidirectional	AT command interface
OLED to ESP32-S3	I2C (SDA/SCL)	MCU to display	3.3V, address 0x3C
Relay driver to compressor	GPIO + optocoupler	MCU to load	Isolated drive via PC817
Buzzer	GPIO (PWM capable)	MCU to output	Active or passive buzzer

3. Power Architecture

The integration system draws all its power from the oxygen concentrator's existing AC supply. A dedicated 12V AC-to-DC module converts mains AC to a stable 12V DC rail. This 12V rail feeds the two purity sensors directly and also supplies an LM2596 buck converter, which steps the voltage down to a regulated 5V for the rest of the system.

3.1 The 12V Rail

The 12V rail supplies only the two OCS-3FL2.0 sensors. These sensors accept 5V to 12V on their power pin, but 12V gives a clean, well-regulated supply that is available directly from the AC-DC module without an extra conversion stage.

3.2 The LM2596 Buck Converter (5V Rail)

The LM2596 is a switching step-down (buck) regulator. It takes the 12V input and produces a stable 5V output capable of supplying up to 3A continuous current. This 5V rail powers the ESP32-S3 DevKit (via its 5V/VIN pin), the SIM800L GSM module (through a voltage drop circuit described below), the OLED display (through the ESP32-S3 onboard 3.3V LDO), the buzzer, and the relay coil drive circuit.

3.3 SIM800L Supply (Voltage Drop Circuit)

The SIM800L operates at 3.4V to 4.4V. It must NOT be connected directly to the 5V rail, as this exceeds its maximum supply voltage. To bring 5V down to the required range, a 1N4001 rectifier diode is placed in series with the SIM800L VCC pin. A silicon diode drops approximately 0.6V to 0.7V in forward bias, so the voltage presented to SIM800L becomes approximately 4.3V to 4.4V, which is within its safe operating window.

A 1000 uF electrolytic capacitor is placed across the SIM800L VCC and GND pins. This is critical. The SIM800L draws burst currents of up to 2A when transmitting GSM signals. Without a bulk capacitor, these current spikes cause the supply voltage to collapse momentarily, crashing the module or causing reset loops. The 1000 uF capacitor acts as a local charge reservoir, absorbing these transient demands.

NOTE: Always install the 1000 uF capacitor as physically close to the SIM800L VCC and GND pins as possible. Long traces between the capacitor and the module reduce its effectiveness during current spikes.

Rail	Voltage	Source	Consumers
12V DC	12V	AC-DC module	OCS-3FL2.0 Sensor 1, Sensor 2
5V DC	5V	LM2596 from 12V	ESP32-S3, Relay circuit, Buzzer, SIM800L drop circuit
SIM800L supply	~4.3V	5V via 1N4001 diode	SIM800L VCC only
3.3V DC	3.3V	ESP32-S3 onboard LDO	OLED, Sensor UART (logic level)

4. Component Descriptions

This section describes every component in the integration module, its role in the system, and relevant operating parameters. Cross-reference with the Bill of Materials (BOM) provided separately.

4.1 ESP32-S3 DevKitC-1

The ESP32-S3 is the central microcontroller of the integration system. It was chosen over the standard ESP32 because this project requires three independent hardware UART channels simultaneously (two for the oxygen sensors and one for the SIM800L), while still keeping UART0 free for USB debug output. The standard ESP32 has only three UARTs total, leaving no comfortable margin. The S3 variant provides the same peripheral set with more GPIO flexibility.

The ESP32-S3 DevKitC-1 board includes a USB-to-serial bridge for programming, a 3.3V LDO regulator, and all GPIO broken out to 0.1-inch headers. It runs at 240 MHz, has 512 KB of SRAM, and supports Wi-Fi and Bluetooth (neither of which are used in this project). Programming is done via the Arduino framework or ESP-IDF using a USB cable.

4.2 LM2596 Buck Converter Module

The LM2596 is a switching voltage regulator configured as a step-down (buck) converter. The pre-built module used in this project includes the LM2596 IC, an inductor, output capacitor, and a potentiometer for adjusting the output voltage. Before installation, the output must be adjusted to exactly 5.0V using a multimeter with no load connected. The LM2596 can supply up to 3A, which is sufficient for all 5V loads in this system combined.

4.3 12V AC-to-DC Power Module

This module converts the mains AC voltage from the oxygen concentrator into a regulated 12V DC supply. It is tapped from the concentrator's internal AC wiring before the compressor switch, so the integration system receives power whenever the concentrator is plugged in, regardless of relay state.

4.4 0.96-inch OLED Display (I2C)

A 128x64 pixel monochrome OLED display connected via I2C. It operates at 3.3V and communicates using SDA and SCL lines. The I2C address is typically 0x3C (verify with an I2C scanner if the display does not initialise). The display shows sensor 1 oxygen percentage, sensor 2 oxygen percentage, current alert status, GSM signal strength (RSSI), and system uptime in seconds or minutes.

4.5 Active Buzzer

An active buzzer contains an internal oscillator and produces a tone when DC voltage is applied. It is driven from a GPIO pin through a current-limiting arrangement. The buzzer pattern varies by alert level: a short, intermittent beep for the first threshold (85%), and a persistent, rapid pattern for the second threshold (70%). The firmware uses PWM or GPIO toggling to generate these patterns.

4.6 1N4001 Rectifier Diode

A general-purpose silicon rectifier diode used in series with the SIM800L power supply line. In forward bias, it drops approximately 0.6V to 0.7V, reducing the 5V rail to approximately 4.3V to 4.4V, which is within the SIM800L safe operating range. The 1N4001 is rated for 1A continuous current, which is sufficient for average SIM800L consumption, though the 1000 uF capacitor handles the transient burst demand.

4.7 1000 uF Electrolytic Capacitor

Placed directly across SIM800L VCC and GND. Provides bulk capacitance to absorb the high current spikes (up to 2A peak) that the SIM800L draws during GSM transmission bursts. Without this capacitor, the supply rail collapses during transmission, causing the module to reset. Polarity must be observed: the positive lead connects to VCC, the negative lead to GND.

5. OCS-3FL2.0 Oxygen Purity Sensor

The OCS-3FL2.0 (WinPower Ultrasonic Oxygen Sensor, 2026 model) is an industrial-grade sensor designed specifically for PSA oxygen concentrators. It measures oxygen concentration, gas flow rate, and gas temperature simultaneously, transmitting all three values over a single UART interface every 500 ms. Two of these sensors are used in this project: one installed before the filtration stage and one after.

5.1 Key Specifications

Parameter	Value
Concentration range	21% to 95.6% O2
Concentration resolution	0.1%
Concentration accuracy	+/- 1.5% FS at 5 to 55 degrees C
Flow range	0 to 20 L/min
Flow resolution	0.1 L/min
Flow accuracy	+/- 0.2 L/min or 5% reading (whichever is larger)
UART output interval	Every 500 ms (automatic reporting mode)
UART baud rate	9600 bps
UART logic level	3.3V TTL (direct connection to ESP32-S3, no level shifting needed)
Warm-up time	10 seconds to specified accuracy
Power supply	DC 5V to 12V, 50 mA
Working temperature	5 to 55 degrees C
Working humidity	5% to 85% RH (no condensation)
Gas sample requirement	No condensation, dust filtration less than 1 um
Connector	J2 socket, 4-pin

5.2 Connector Pinout (J2 Socket)

Pin	Function	Notes
Pin 1	VCC (Power input)	Connect to 12V or 5V DC. 12V used in this project.

Pin 2	RXD	Data input to sensor (from ESP32-S3 TX pin)
Pin 3	TXD	Data output from sensor (to ESP32-S3 RX pin)
Pin 4	GND	Common ground

NOTE: Sensor TXD connects to the ESP32-S3 RX pin, and sensor RXD connects to the ESP32-S3 TX pin. This cross-connection is correct and intentional. A common wiring mistake is connecting TX to TX and RX to RX, which results in no data.

5.3 Placement in the System

Sensor 1 is placed on the raw oxygen outlet of the concentrator, before the primary filtration or molecular sieve output. Sensor 2 is placed on the final oxygen outlet tube, after all filtration stages. By comparing the two readings, the system can identify whether a drop in purity is occurring at the filtration stage or further downstream. Both sensors must be installed in-line with the oxygen tube, following the arrow marked on the sensor body that indicates gas flow direction.

6. UART Communication and Sensor Protocol

UART (Universal Asynchronous Receiver/Transmitter) is a serial communication protocol that transmits data one bit at a time over two wires: TX (transmit) and RX (receive). Both devices must agree on the same baud rate (number of bits per second) for communication to work correctly. The OCS-3FL2.0 sensor uses 9600 bps, 8 data bits, 1 stop bit, and no parity check.

6.1 Sensor Operating Modes

The sensor operates in two modes. In **automatic reporting mode**, it transmits a data packet every 500 ms without any command from the microcontroller. This mode activates automatically if the sensor does not receive a query command within 4 seconds. In **query/response mode**, the microcontroller sends a 4-byte query command and the sensor responds immediately with a data packet. This project uses automatic reporting mode for simplicity, meaning the ESP32-S3 simply listens on each UART channel and parses incoming packets.

6.2 Data Packet Structure

Every data packet transmitted by the sensor is 9 bytes long, all in hexadecimal. The packet structure is fixed and must be validated before the data is used. Validation requires checking that the first three bytes match the expected header values, and that the checksum (last byte) equals the sum of all preceding bytes, truncated to 8 bits.

Byte Index	Hex Value	Field	Description
Byte 0	0x16	Header 1	Fixed identifier, must equal 0x16
Byte 1	0x09	Header 2	Fixed identifier, must equal 0x09
Byte 2	0x01	Header 3	Fixed identifier, must equal 0x01
Bytes 3-4	--	O2 Concentration	16-bit value: (Byte3 x 256 + Byte4) / 10 = % O2
Bytes 5-6	--	Flow Rate	16-bit value: (Byte5 x 256 + Byte6) / 10 = L/min
Bytes 7-8	--	Temperature	16-bit value: (Byte7 x 256 + Byte8) / 10 = degrees C
Byte 8 (last)	--	Checksum (CS)	Sum of bytes 0 through 7, modulo 256

6.3 Packet Parsing Example (C/Arduino style)

```
uint8_t buf[9];  
  
// Read 9 bytes from UART into buf[]
```

```
// Step 1: Validate header
if (buf[0]==0x16 && buf[1]==0x09 && buf[2]==0x01) {

// Step 2: Validate checksum
uint8_t cs = 0;
for (int i = 0; i < 8; i++) cs += buf[i];
if (cs == buf[8]) {

// Step 3: Extract values
float o2_pct = (buf[3]*256 + buf[4]) / 10.0; // Oxygen %
float flow = (buf[5]*256 + buf[6]) / 10.0; // L/min
float temp_c = (buf[7]*256 + buf[8]) / 10.0; // degrees C
}
}
```

Note: The checksum in the packet description uses byte index 8 for both temperature high byte and checksum. In the actual 9-byte packet, bytes 7 and 8 are temperature high and low, and byte 8 (the 9th byte) is the checksum. Always refer to the manufacturer datasheet for exact byte count confirmation during implementation.

7. Relay Driver Circuit

The relay switches the compressor power supply on or off based on the oxygen purity level. Because the relay coil is connected to mains-voltage-switching contacts, it is essential that the ESP32-S3 microcontroller is electrically isolated from the relay and load circuit. This isolation is achieved using a two-stage driver: an optocoupler (PC817) followed by an NPN transistor (BC337).

7.1 Circuit Topology and Signal Flow

Stage 1 - ESP32-S3 GPIO Output

The ESP32-S3 drives a GPIO pin HIGH (3.3V) to activate the relay, or LOW (0V) to deactivate it. The GPIO never connects directly to any component that touches the relay coil side.

Stage 2 - Current Limiting Resistor (220 ohm)

The GPIO output connects through a 220 ohm resistor to the anode of the PC817 input LED. The PC817 input LED has a forward voltage of approximately 1.2V. With 3.3V on the GPIO: $I = (3.3 - 1.2) / 220 = 9.5$ mA. This is within the PC817 recommended input current range of 5 mA to 50 mA.

Stage 3 - PC817 Optocoupler

The PC817 is an optocoupler, meaning it transmits signals using light rather than a direct electrical connection. Inside the package, an infrared LED (input side) shines on a phototransistor (output side). When the input LED conducts, the phototransistor turns on, creating a path from its collector to emitter. The input and output circuits share no electrical connection whatsoever. This is the isolation barrier that protects the ESP32-S3.

Stage 4 - BC337 NPN Transistor

The PC817 phototransistor output drives the base of the BC337 NPN transistor. The BC337 is a small-signal NPN transistor rated for collector currents up to 800 mA. When the PC817 output pulls the BC337 base high, the transistor saturates and connects the relay coil to ground, energising the relay.

Stage 5 - Pull-down Resistor (10k ohm on BC337 base)

A 10k ohm resistor is connected from the BC337 base to GND. This ensures the transistor base is held firmly at 0V when the PC817 output is off, preventing the relay from being triggered by noise or floating inputs. This is especially important during ESP32-S3 boot-up, when GPIO states are briefly undefined.

Stage 6 - Flyback Diode (1N4001 or 1N4007)

The relay coil is an inductive load. When the BC337 switches off and the coil current is interrupted, the magnetic field collapses and generates a large reverse voltage spike (potentially hundreds of volts for a brief instant). The flyback diode is placed across the relay coil terminals in reverse bias during normal operation. When the voltage spike occurs and reverses polarity, the diode becomes forward biased and clamps the spike, dissipating it harmlessly. Without this diode, the spike can destroy the BC337 transistor.

DANGER: The relay contacts switch mains AC voltage. Never touch relay output terminals or the compressor wiring while the system is powered. Treat the relay output side as live mains voltage at all times.

7.2 Component Summary for Relay Driver

Component	Value/Part	Role
Resistor R1	220 ohm	Limits current through PC817 input LED from ESP32-S3 GPIO
Optocoupler	PC817	Electrical isolation between ESP32-S3 and relay driver circuit
Transistor Q1	BC337 NPN	Switches relay coil current to GND when activated
Resistor R2	10k ohm	Pull-down on BC337 base, prevents false triggering at boot
Diode D1	1N4001 or 1N4007	Flyback protection across relay coil against inductive spikes
Relay	5V coil (TBD spec)	Switches compressor AC load

8. SIM800L GSM Module

The SIM800L is a compact GSM/GPRS module that provides cellular communication capability. In this project, it is used exclusively for SMS (short message service) and voice call initiation. It communicates with the ESP32-S3 using AT commands over UART at 9600 bps.

8.1 Network Compatibility

The SIM800L supports GSM 850/900/1800/1900 MHz quad-band. Nigerian GSM networks (MTN, Airtel, Glo, 9mobile) operate on the 900 MHz and 1800 MHz bands, which are both covered. Any active Nigerian SIM card with sufficient airtime credit for SMS and calls will work. The SIM card is inserted directly into the SIM800L module holder with the chip facing down.

8.2 Key AT Commands Used

AT Command	Function
AT	Basic connectivity check. Module responds with OK.
AT+CSQ	Query signal strength (RSSI). Returns value 0 to 31. Display on OLED.
AT+CMGF=1	Set SMS mode to text (not PDU). Required before sending SMS.
AT+CMGS="+234XXXXXXXXXX"	Send SMS. Follow with message text and Ctrl+Z (0x1A) to send.
ATD+234XXXXXXXXXX;	Place a voice call. The semicolon specifies a voice call, not data.
ATH	Hang up / end a call.
AT+CREG?	Check network registration status. Should return +CREG: 0,1 when registered.

NOTE: The SIM800L TX and RX logic level is 2.8V. The ESP32-S3 outputs 3.3V on its TX pin. A voltage divider (1k and 2k resistors) on the line from ESP32-S3 TX to SIM800L RX is recommended to bring 3.3V down to approximately 2.2V, which is within the SIM800L input threshold. The SIM800L TX to ESP32-S3 RX connection is direct (2.8V is read correctly as HIGH by the ESP32-S3 3.3V logic input).

9. Alert Logic and Threshold Behaviour

The alert system monitors the oxygen purity reading from Sensor 2 (post-filtration output). This is the value that represents what the patient is actually receiving. Sensor 1 (pre-filtration) is used for comparative monitoring and displayed on the OLED, but the thresholds and relay control decisions are based on Sensor 2.

9.1 Threshold Definitions

Level	Threshold	Condition	System Response
Normal	Above 85%	Purity is within safe range	No alert. Compressor runs. OLED shows green status.
Warning (Level 1)	70% to 85%	Purity is below safe minimum	Buzzer sounds intermittent short beeps. SMS sent to caregiver. Compressor continues.
Danger (Level 2)	Below 70%	Purity is critically low, risk to patient	Buzzer sounds rapid, persistent pattern. Voice call placed to caregiver. Relay cuts compressor.

9.2 State Machine Description

The firmware implements a simple three-state machine: NORMAL, WARNING, and DANGER. Transitions between states are based on the most recent valid Sensor 2 reading. To prevent false triggers from a single bad packet, a debounce count is used: the system only transitions to a higher alert state after receiving a set number of consecutive readings below the threshold (recommended: 3 consecutive readings). It transitions back to a lower state immediately when readings recover, but the buzzer and compressor are only reset after sustained recovery.

9.3 SMS Alert Behaviour

When the system enters WARNING state, one SMS is sent to the caregiver number. The message includes the current O2 percentage from Sensor 2 and the system uptime. The SMS is not repeated unless the system recovers to NORMAL state and then drops to WARNING again. This prevents flooding the caregiver with repeated messages.

Example SMS content: 'ALERT: Oxygen purity low. Sensor reading: 82%. System uptime: 04:32 min. Please check the concentrator.'

9.4 Voice Call Behaviour

When the system enters DANGER state, a voice call is placed to the caregiver number. The call is held open for a configurable duration (e.g. 30 seconds) to allow the caregiver to hear the alert tone or respond. The call is then terminated with ATH. The relay simultaneously opens, cutting power to the compressor. A retry mechanism attempts the call again if the first attempt fails (no answer or network error).

9.5 Relay and Compressor Control

In NORMAL and WARNING states, the relay is energised (closed), allowing the compressor to run. In DANGER state, the relay is de-energised (open), cutting compressor power. Note that the relay is wired in normally-open configuration for this purpose: the relay must be actively held closed by the ESP32-S3 during normal operation. This is a fail-safe design choice: if the ESP32-S3 loses power or crashes, the relay opens and the compressor stops, rather than continuing to run unmonitored.

DANGER: Relay wiring must follow normally-open (NO) configuration as described above. Connecting to the normally-closed (NC) terminal inverts the safety logic and will cause the compressor to run even when the ESP32-S3 is off.

10. OLED Display

The 0.96-inch 128x64 pixel OLED display provides a local real-time status view without requiring any phone or external device. It is driven over I2C using the SSD1306 or SH1106 controller (verify your specific module). Power is drawn from the ESP32-S3's onboard 3.3V rail.

Display Area	Content Shown
Line 1	S1: XX.X% S2: XX.X% (both sensor O2 readings)
Line 2	Flow: XX.X L/min Temp: XX.X C
Line 3	Status: NORMAL / WARNING / DANGER
Line 4	GSM: XX dBm SIM: OK / FAIL
Line 5	Uptime: XXX sec or XX:XX min

11. Caregiver Number Configuration

The caregiver phone number is not hardcoded in the firmware. It is stored in the ESP32-S3 non-volatile storage (NVS flash), which retains data even after power loss. Two methods are available to set or update the number:

Method A - USB Serial (Initial Setup)

Connect the ESP32-S3 to a computer via USB. Open any serial monitor (Arduino IDE, PuTTY, or similar) at 115200 baud. The system will print a menu on startup. Type the caregiver number in international format (e.g. +2348012345678) and press Enter. The number is saved to NVS and the menu exits. This is the recommended method for initial configuration before deployment.

Method B - SMS Command (Field Update)

To change the number without physical access to the device, send an SMS from any phone to the SIM card installed in the SIM800L. The SMS must follow this exact format:

```
SET : +2348099887766
```

The ESP32-S3 parses incoming SMS messages continuously. When it detects a message starting with SET:, it extracts the number that follows, validates it, saves it to NVS, and sends a confirmation SMS back to the sender. The old number is overwritten.

NOTE: The SET: command is case-sensitive. The number must include the country code in international format starting with +. Example: +2348012345678 for a Nigerian MTN number.

12. ESP32-S3 Pin Assignment

The following table shows the proposed GPIO pin assignments for the ESP32-S3 DevKitC-1. Pin assignments are subject to revision when the full schematic is completed in Phase 3. GPIO numbers shown are ESP32-S3 internal GPIO numbers, not physical pin numbers on the board header.

GPIO	Function	Direction	Connected To
GPIO 17	UART1 TX	Output	Sensor 1 RXD (pre-filter)
GPIO 18	UART1 RX	Input	Sensor 1 TXD (pre-filter)
GPIO 19	UART2 TX	Output	Sensor 2 RXD (post-filter)
GPIO 20	UART2 RX	Input	Sensor 2 TXD (post-filter)
GPIO 43	UART0 TX	Output	SIM800L RX (via voltage divider)
GPIO 44	UART0 RX	Input	SIM800L TX
GPIO 8	I2C SDA	Bidirectional	OLED SDA
GPIO 9	I2C SCL	Output	OLED SCL
GPIO 10	Relay control	Output	PC817 input LED (via 220 ohm)
GPIO 11	Buzzer	Output (PWM)	Active buzzer positive terminal

NOTE: GPIO 43 and 44 are the default UART0 (USB debug) pins on the ESP32-S3 DevKitC-1. If UART0 is reassigned to SIM800L, debug output must move to a different UART or be disabled. Confirm pin mapping against your specific DevKitC-1 board pinout diagram before wiring.

13. Enclosure and Mechanical Integration

All integration electronics are housed in a white project enclosure box placed externally beside the oxygen concentrator. The concentrator itself remains physically unmodified. The following connections pass between the enclosure and the concentrator:

Connection	Type	Description
AC power input	Electrical wire	Tapped from concentrator AC supply to 12V AC-DC module
Compressor relay output	Electrical wire	Relay contact in series with compressor AC supply line
Oxygen sample tube (Sensor 1)	Silicone or PTFE tube	In-line with oxygen path before filtration
Oxygen sample tube (Sensor 2)	Silicone or PTFE tube	In-line with oxygen path after filtration

Panel cutouts on the enclosure are needed for: the OLED display window, the buzzer grille, a SIM card access slot or door, the USB programming port, and cable glands for the oxygen tubes and power/relay wires. The GSM antenna wire for the SIM800L should be routed to exit the enclosure and be positioned away from metal surfaces for best signal reception.

Appendix A - Sensor UART Packet Quick Reference

The OCS-3FL2.0 sensor transmits a 9-byte packet every 500 ms in automatic mode. All values are in hexadecimal. Validate the header (bytes 0-2) and checksum (byte 8) before using the data.

Byte	0	1	2	3	4	5	6	7	8
Field	Header	Header	Header	O2 High	O2 Low	Flow High	Flow Low	Temp	Checksum
Fixed value	0x16	0x09	0x01	--	--	--	--	--	SUM mod 256

O2 concentration (%) = (Byte3 x 256 + Byte4) / 10

Flow rate (L/min) = (Byte5 x 256 + Byte6) / 10

Temperature (deg C) = (Byte7 x 256 + Byte8) / 10

Appendix B - Revision History

Revision	Date	Phase	Changes
Rev 1.0	22 Jun 2026	Phase 1	Initial release. System architecture, power design, sensor protocol, relay driver, alert logic, pin assignment, enclosure overview.
Rev 1.1	TBD	Phase 2	Pictorial wiring diagram added (Circuit Designer).
Rev 1.2	TBD	Phase 3	KiCad schematic inserted. Component-level circuit annotations.
Rev 1.3	TBD	Phase 4	PCB layout image. Single-layer routing notes.